

Software Architecture 2026

Lecture 7, part 1

Architectural tactics and patterns, part 2

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What's an architectural tactic?

Recognizing **tactics** in real life

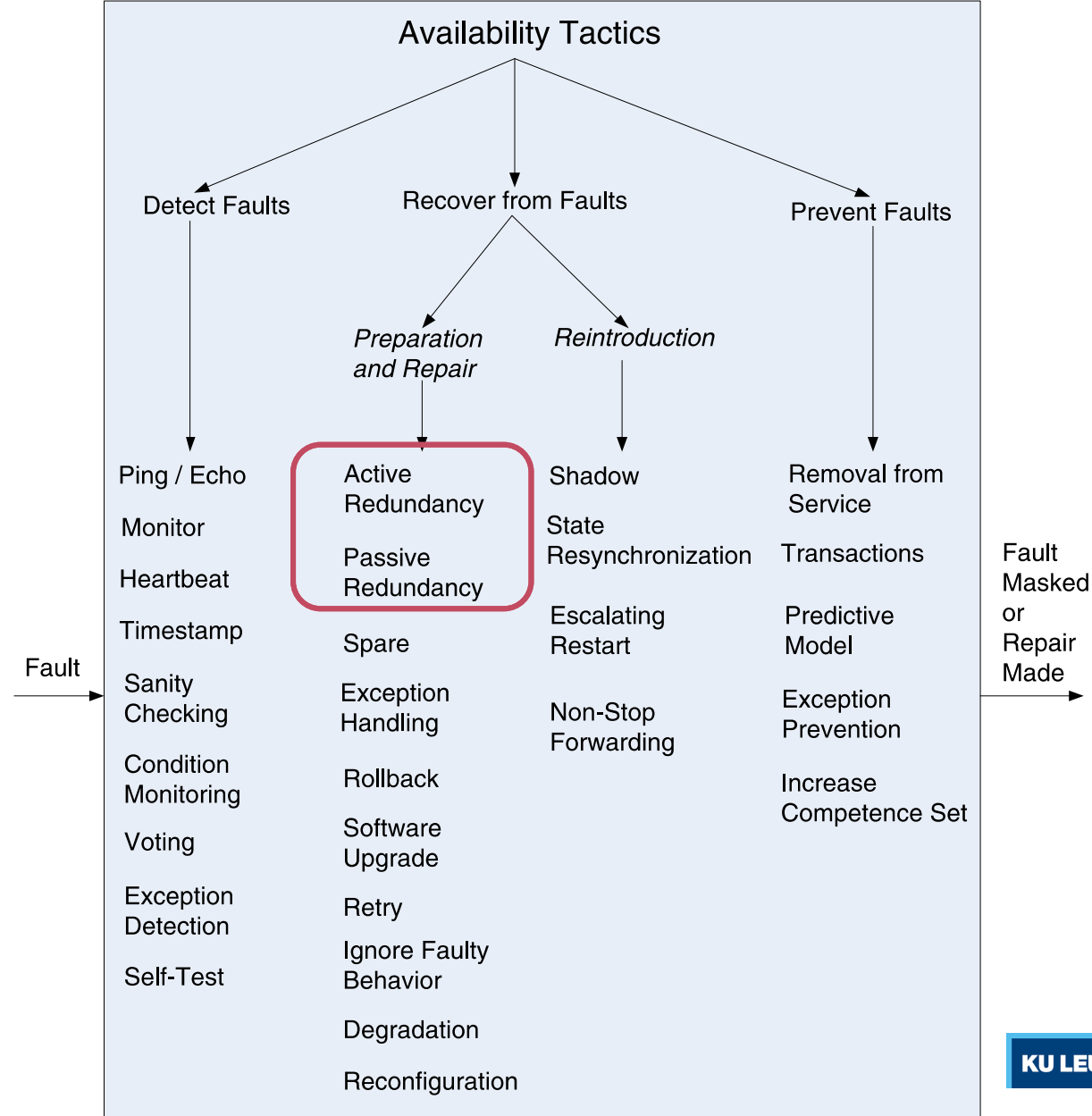
traveling to a conference



Performance tactics (SAiP Ch. 8)



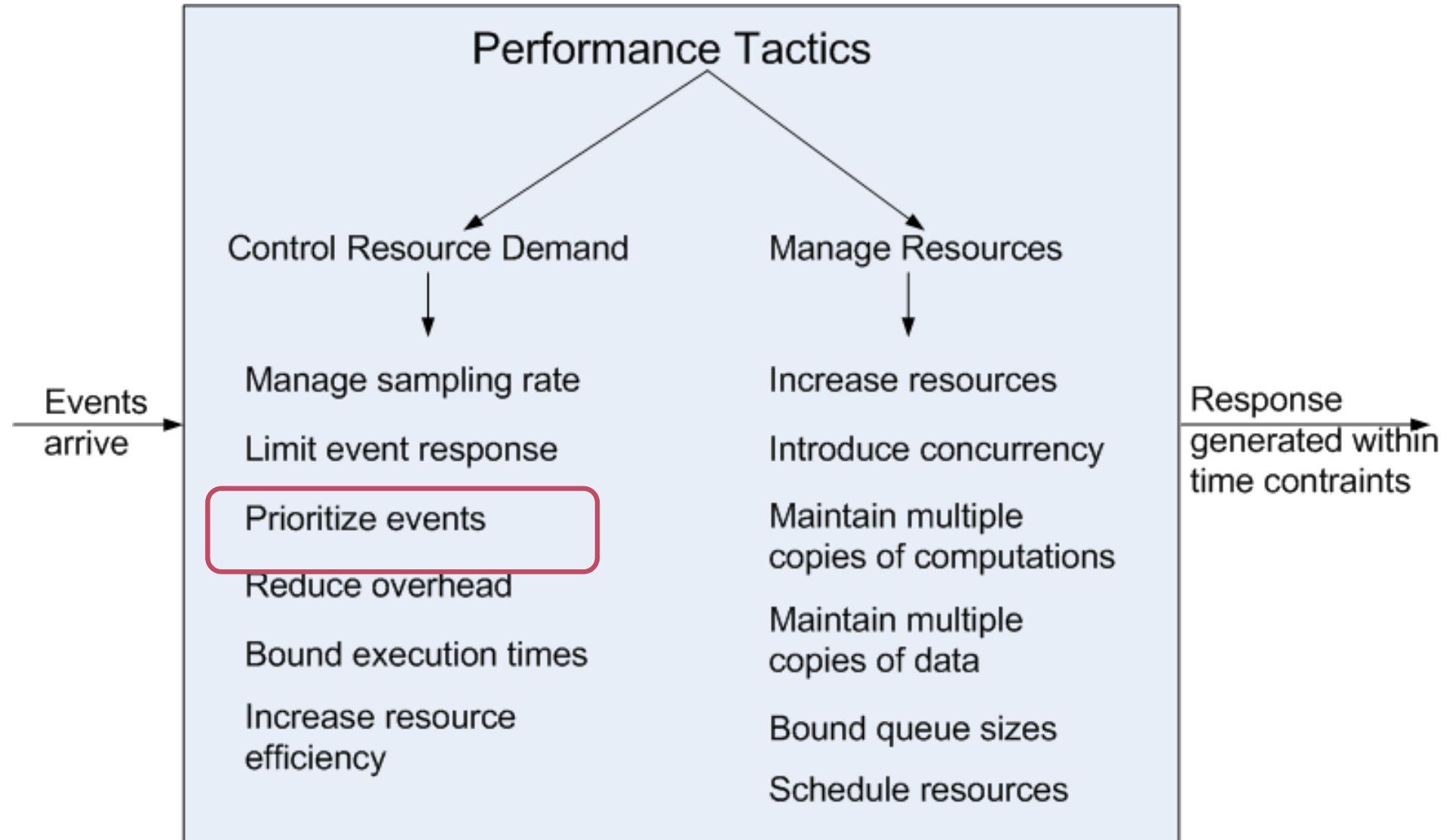
Availability tactics (SAiP Ch. 5)



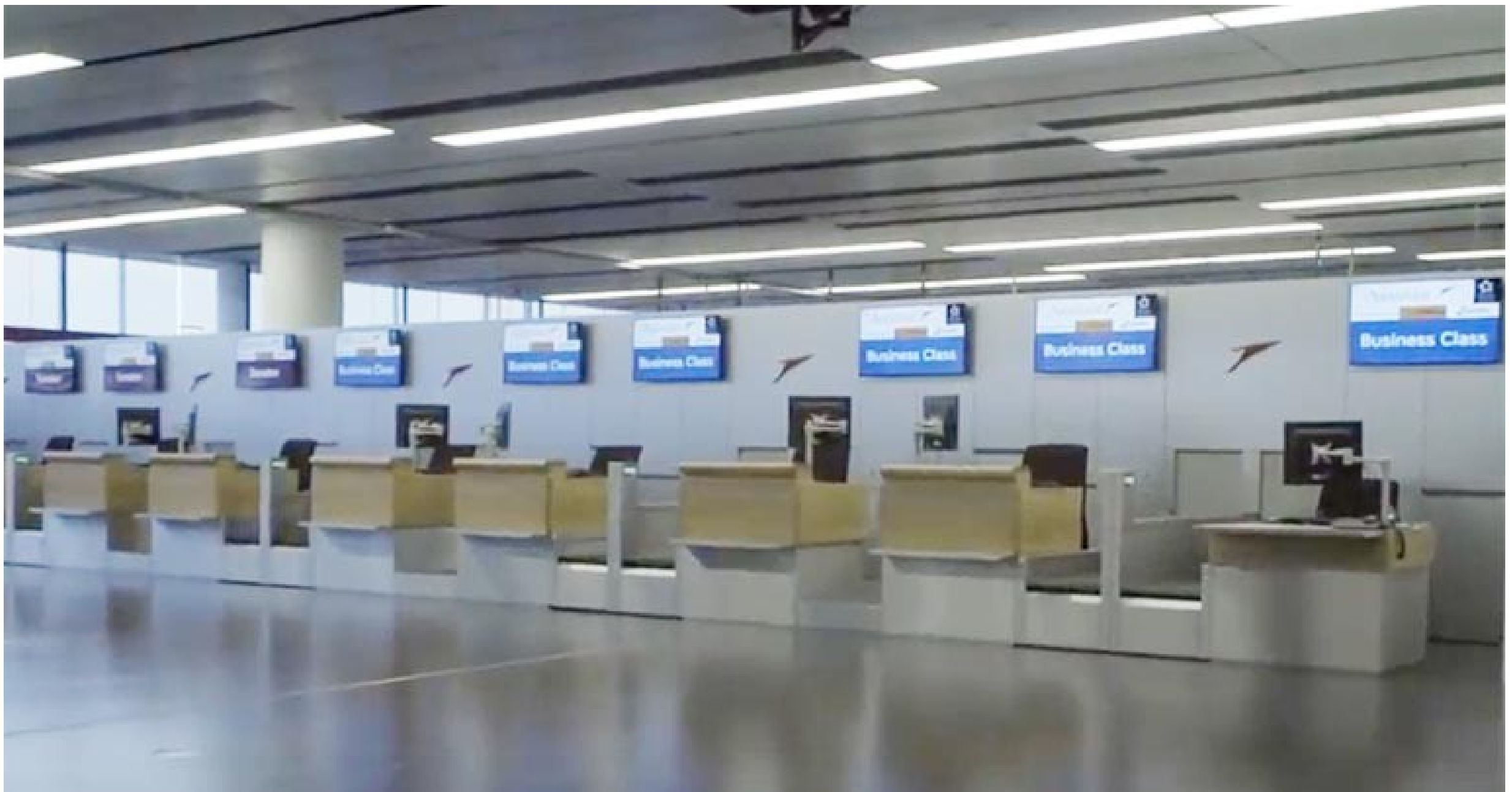




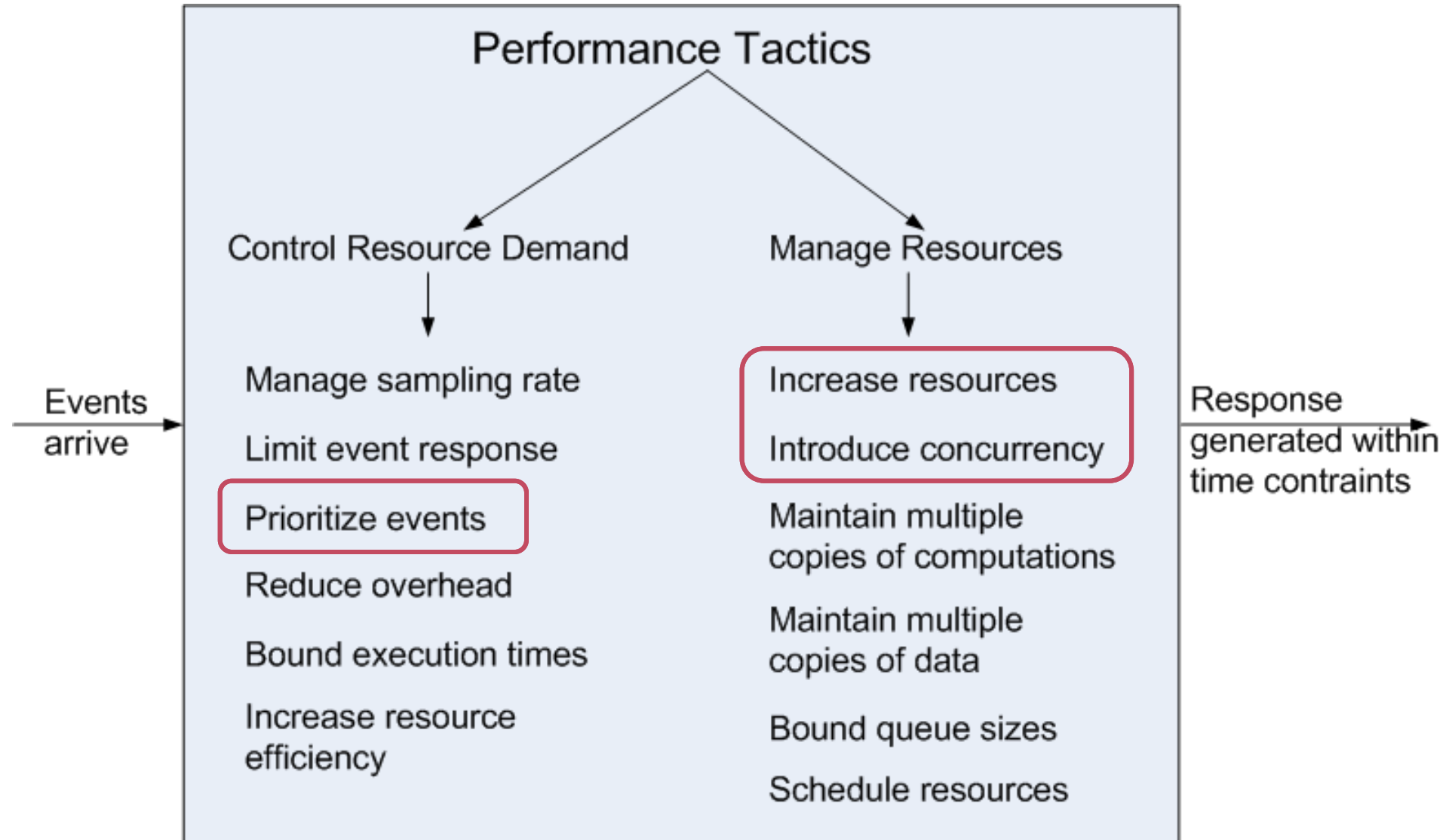
Performance tactics (SAiP Ch. 8)







Performance tactics (SAiP Ch. 8)



Online check-in

Go ahead with your experience, your seat and get your boarding pass

Write the surname, code and origin airport in the booking.

Surname

Booking code

Departure

Check in

Or, if you like, enter your Iberia Plus account to check in.

Iberia Plus Number

PIN

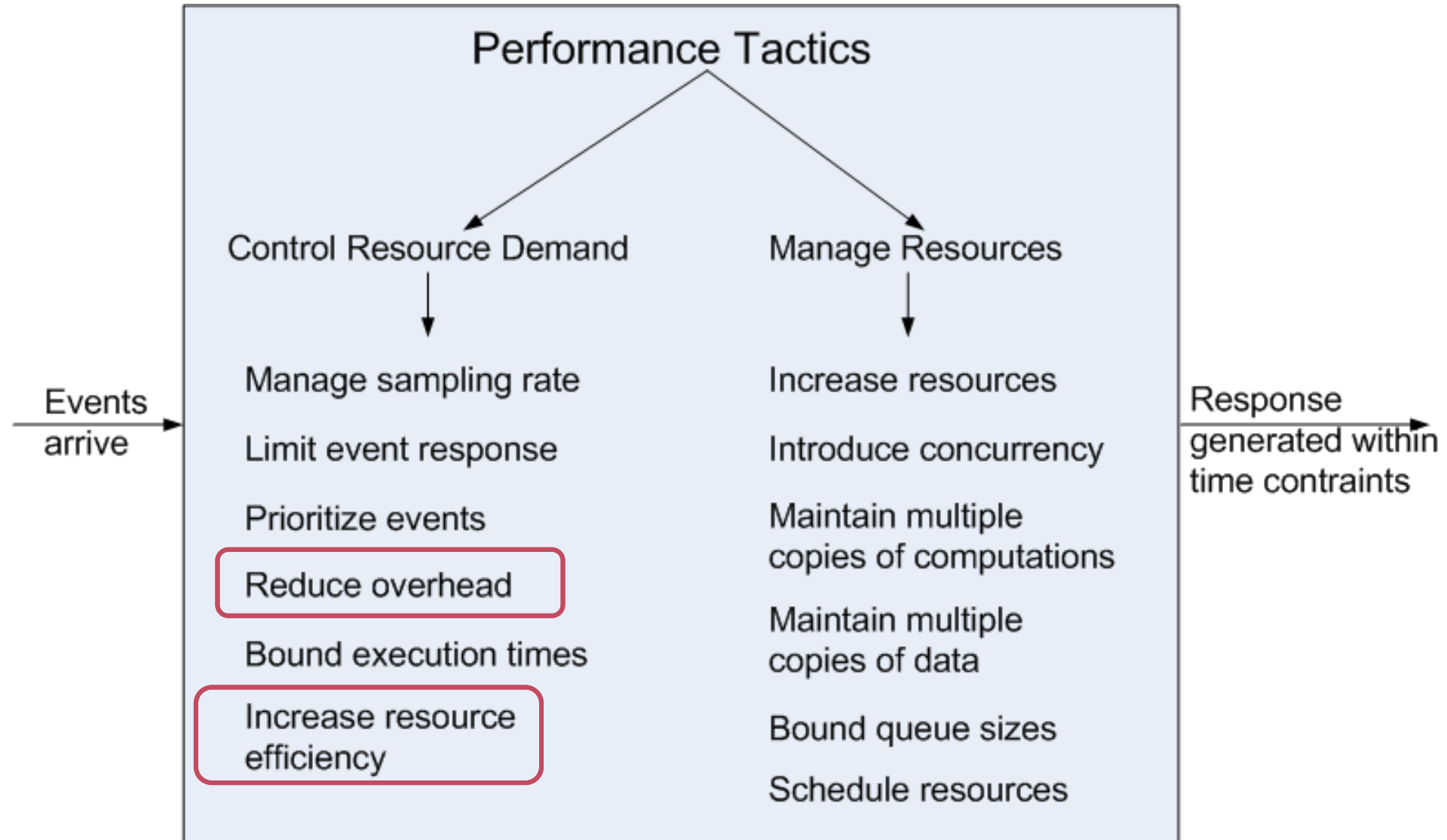
Forgot or do not have
a PIN?

Departure

Start session

Performance tactics (SAiP Ch. 8)

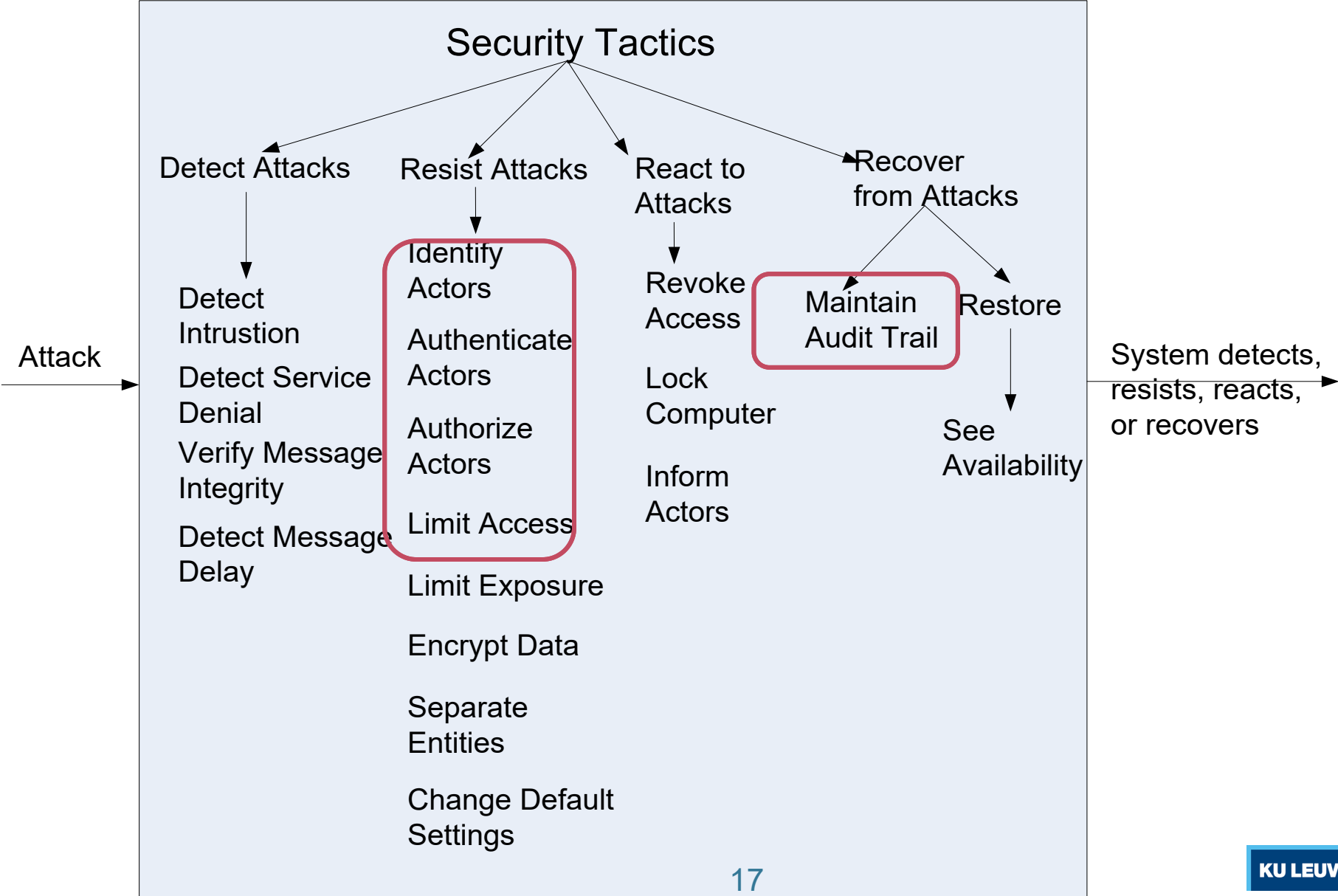


Airport Security





Security tactics (SAiP Ch. 9)



Priority Lane

Fast Track

First Class

ファーストクラス

Business Class

ビジネスクラス

oneworld Emerald & Sapphire

ワンワールド エメラルド&サファイア

Premium Member

プレミアムメンバー

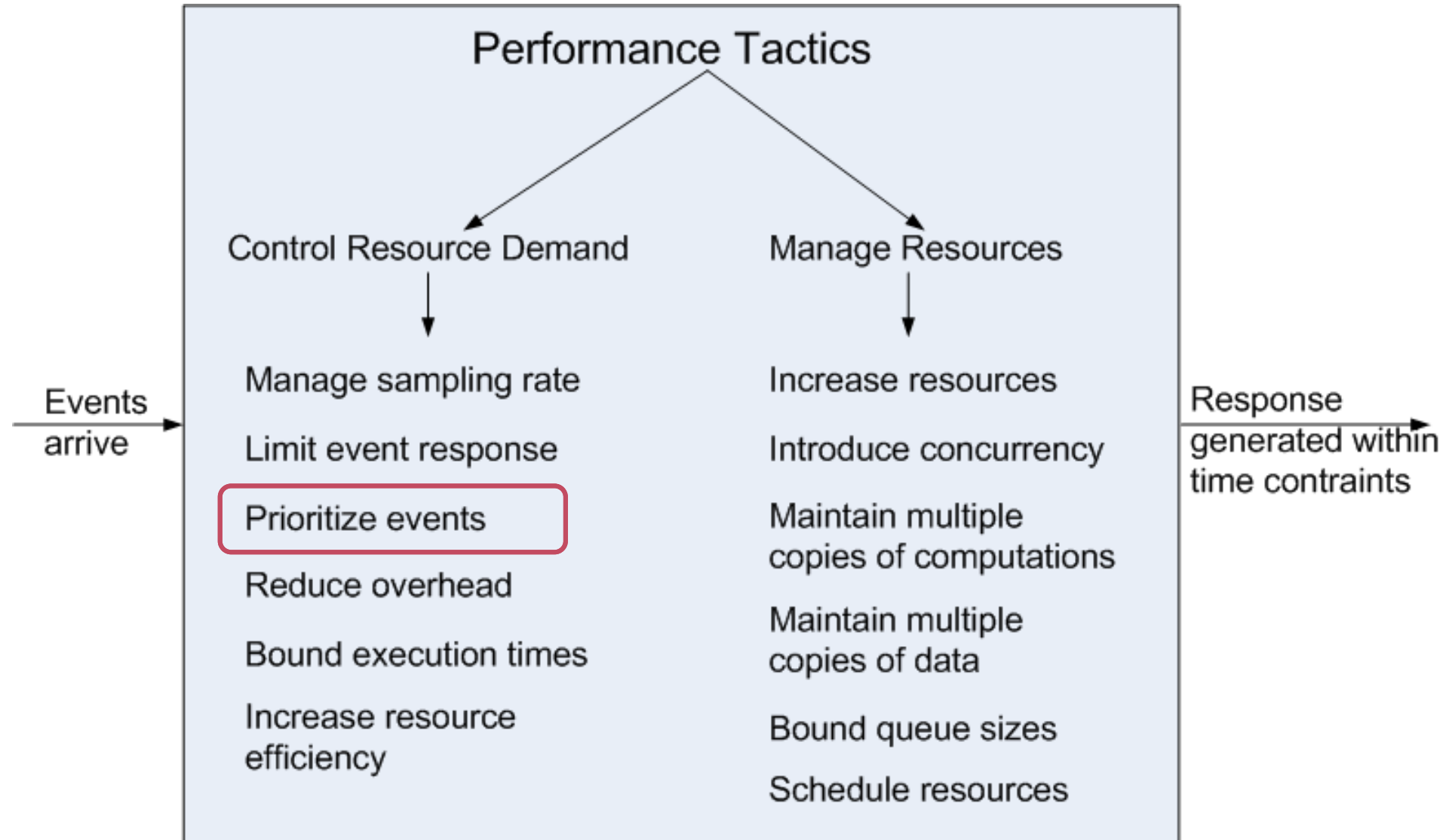


Priority Lane

優先レーン



Performance tactics (SAiP Ch. 8)





Database sharding

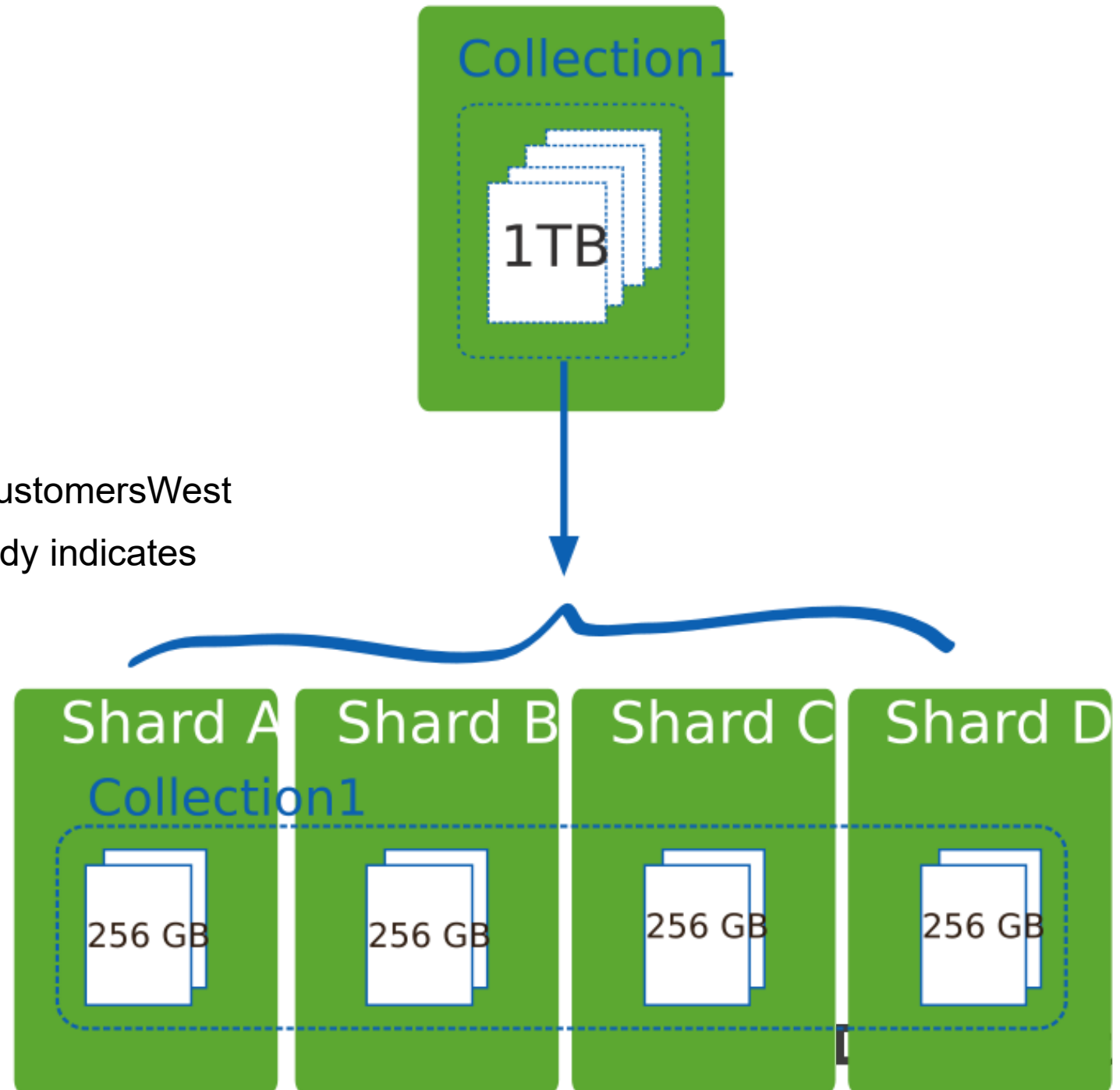
Distributing requests

› Scalability

- › Database writes
- › Shard criterion or key
 - ›› E.g. Shard CustomersEast and CustomersWest tables, where their **zip code** already indicates where they will be found.

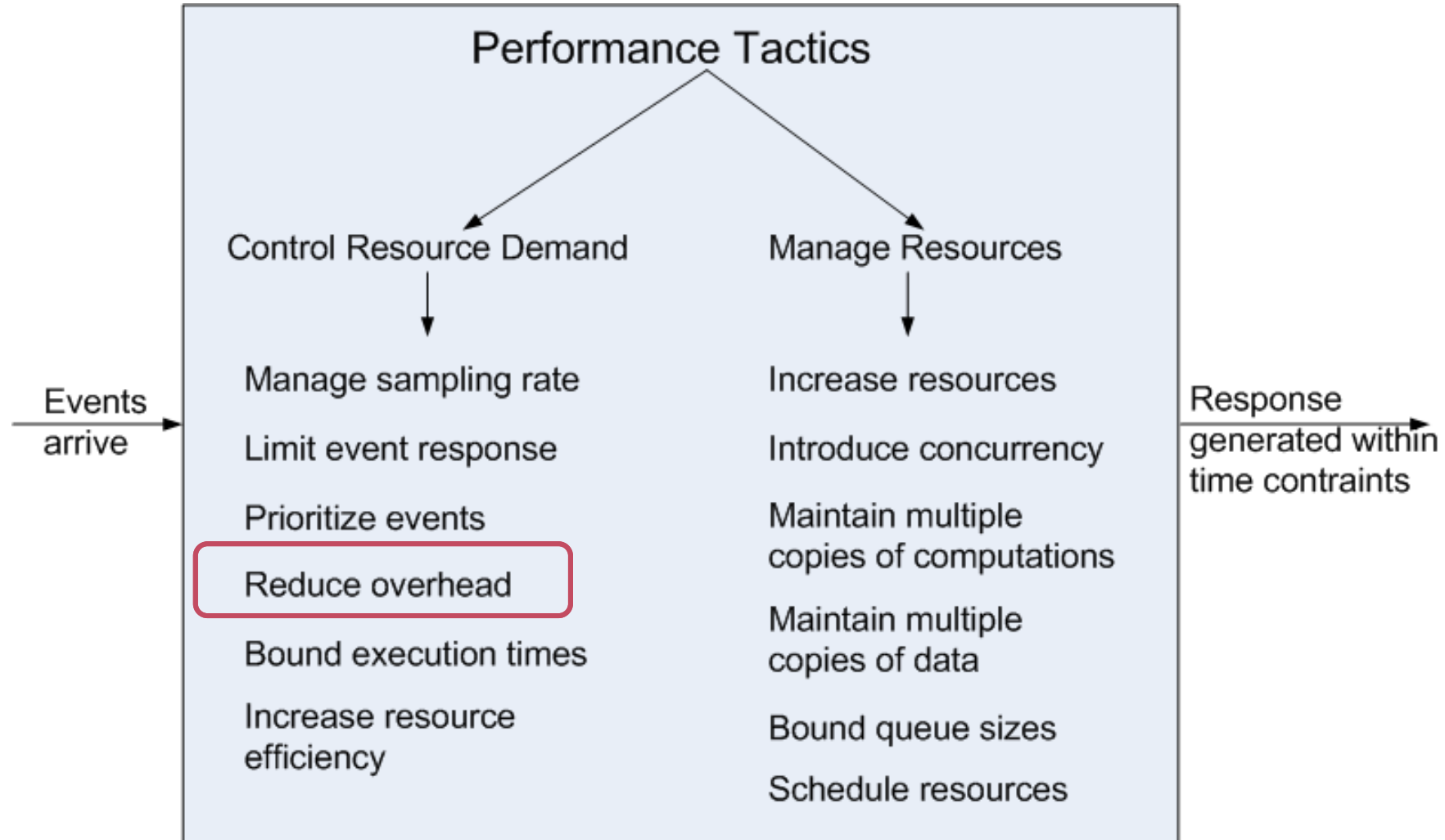
› Downsides:

- › More complicated querying
- › More complicated back-up
- › Single points of failure (Av!) with less impact (only a shard)





Performance tactics (SAiP Ch. 8)

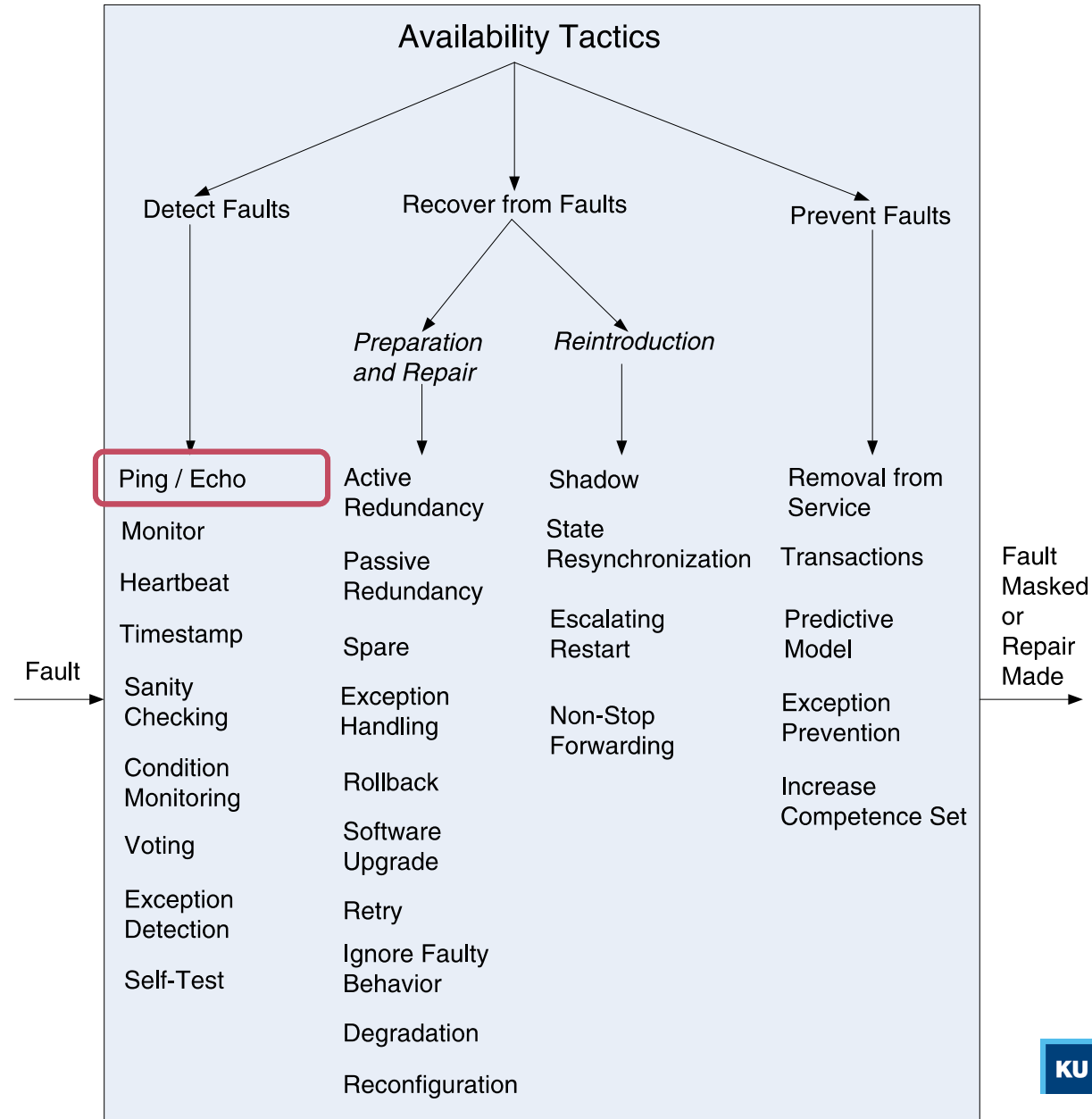


iMessage
Today 7:21 PM

hey

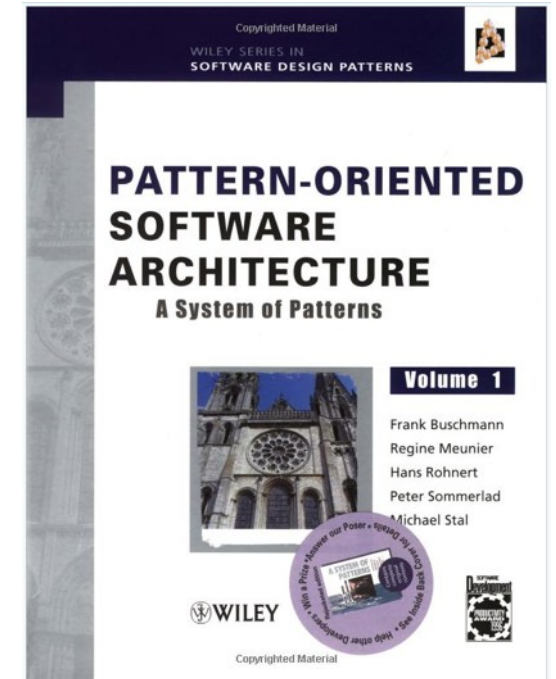
are you okay?

Availability tactics (SAiP Ch. 5)



Introduction to some other architectural patterns and styles

- › Previous lecture:
 - ›› Replicated component group
 - ›› Layering
 - ›› Pipe-and-filter
- › Outline:
 - ›› (multi-)tiered architectures
 - ›› Event-based/event-driven architecture

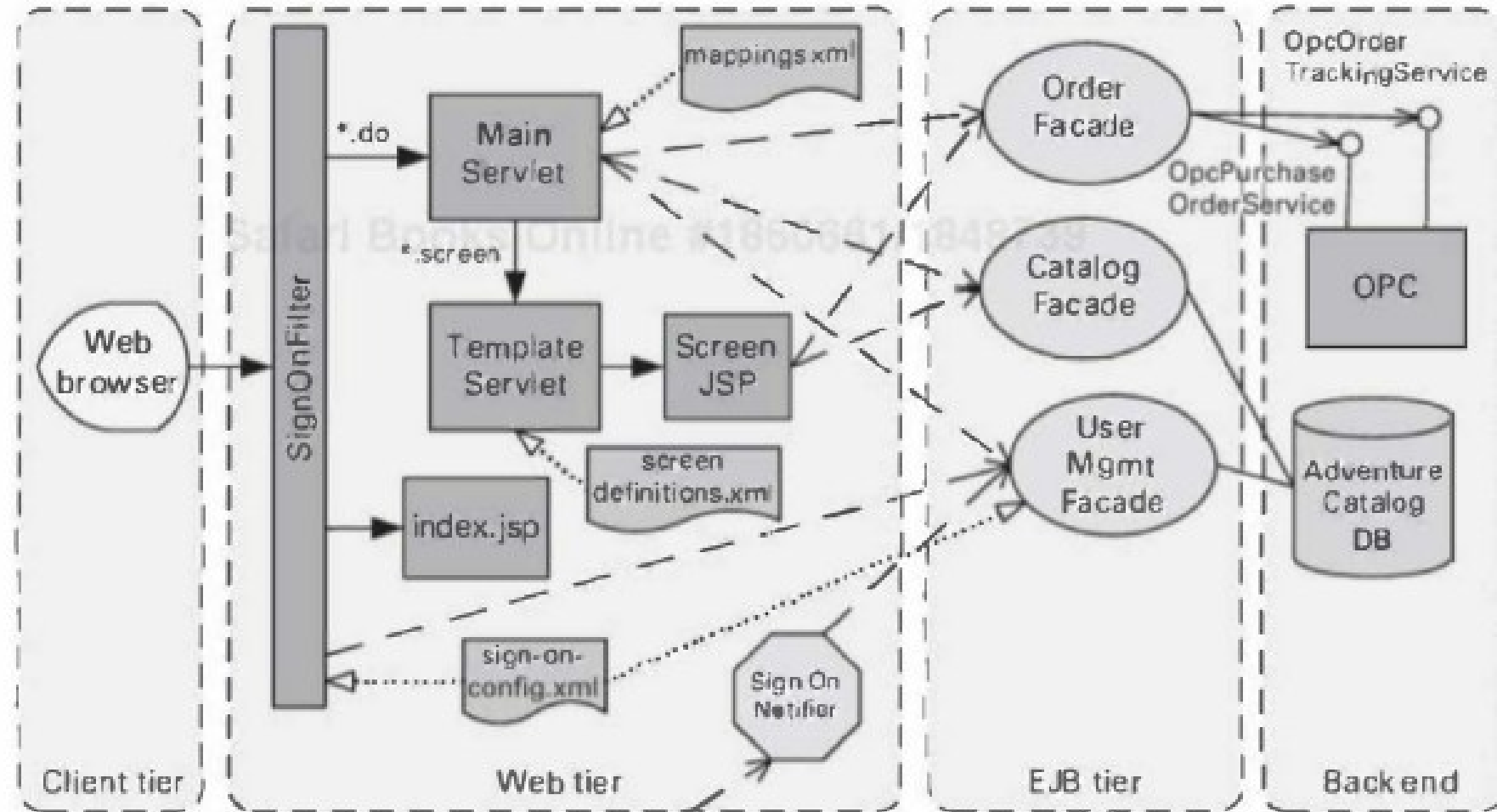


(multi-)tiered architectures

Tiering/Multi-tier

Allocation pattern

- › Components
- › Nodes



Key



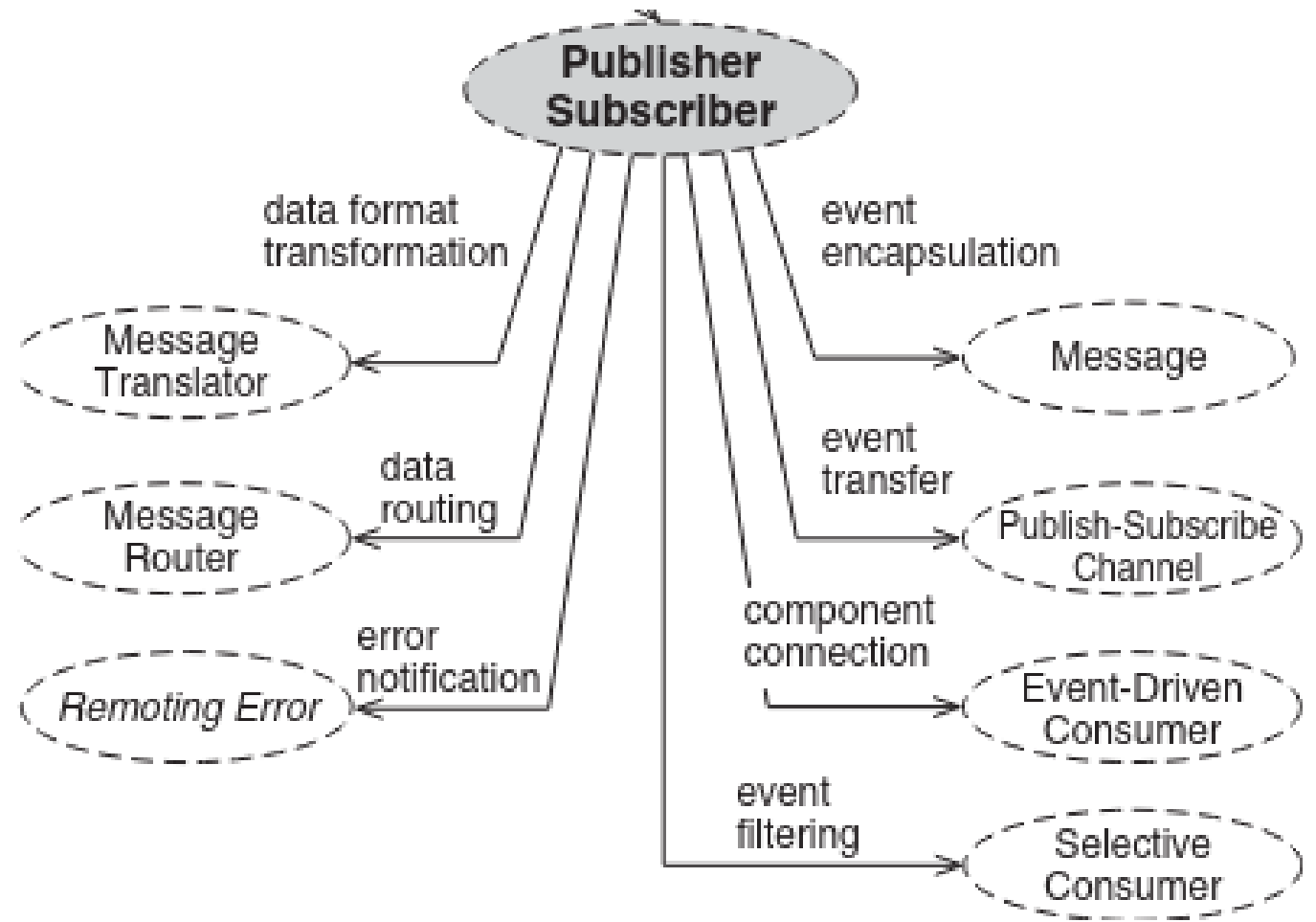
Event-based architecture

Event-based architecture

- › Event is “a significant change in state”
 - ›› Event is created and sent by event source (emitter)
 - ›› Event is propagated to loosely coupled software components (consumers/sinks/handlers)
- › Principles
 - ›› Asynchronous communication: fire-and-forget
 - ›› Loose coupling: source does not know the consumer

Related patterns

- › Publish-subscriber
- › Service-oriented architecture (SOA)

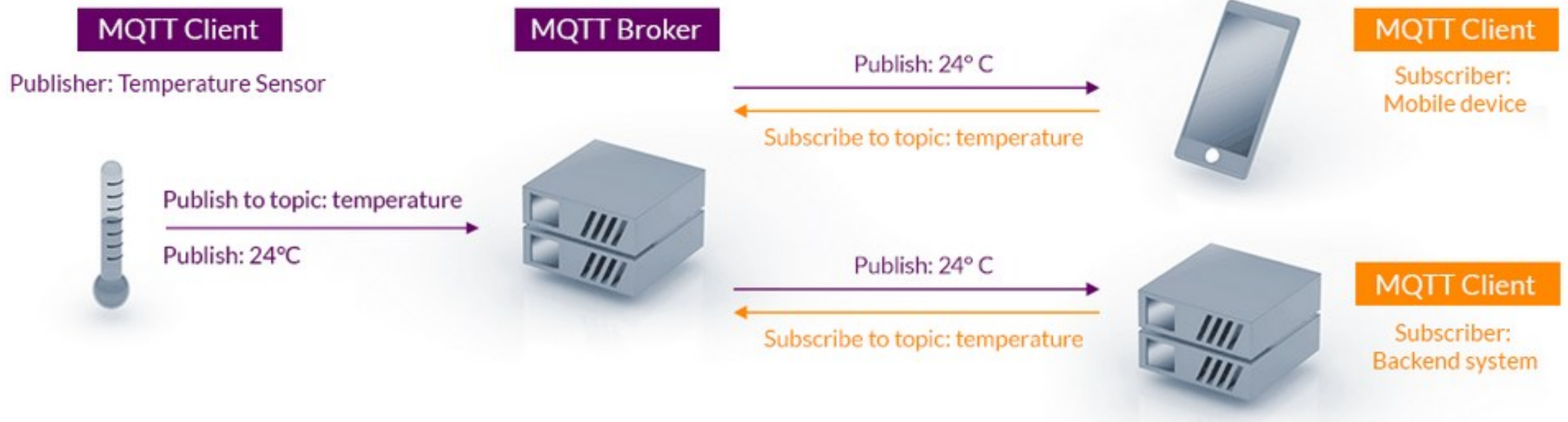


Publish-subscribe

E.g. MQTT for IoT



MQTT Publish / Subscribe Architecture



Client and brokers:

clients publish and subscribe to topics

brokers relay mqtt events

Recap

- › *Architecture-decision making: using common sense and not re-inventing the wheel*
- › *Known solutions to common problems*
 1. Design principles
 2. Architectural tactics
 3. Architectural patterns
 4. Reference architectures / frameworks

POINTERS AND
DEFINITIONS IN
LECTURE 6

Much knowledge can be obtained through study of existing systems,
to recognize key decisions and understanding their rationale